

Training DNNs Resilient to Adversarial and Random Bit-Flips by Learning Quantization Ranges

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Abstract

Promoting robustness in deep neural networks (DNNs) is crucial for their reliable deployment in uncertain environments, such as low-power settings or in the presence of adversarial attacks. In particular, bit-flip weight perturbations in quantized networks can significantly degrade performance, underscoring the need to improve DNN resilience. In this paper, we introduce a training mechanism to learn the quantization range of different DNN layers to enhance DNN robustness against bit-flip errors on the model parameters. The proposed approach, called weight clipping-aware training (WCAT), minimizes the quantization range while preserving performance, striking a balance between the two. Our experimental results on different models and datasets showcase that DNNs trained with WCAT can tolerate a high amount of noise while keeping the accuracy close to the baseline model. Moreover, we show that our method significantly enhances DNN robustness against adversarial bit-flip attacks. Finally, when considering the energy-reliability trade-off inherent in on-chip SRAM memories, we observe that WCAT consistently improves the Pareto frontier of test accuracy and energy consumption across diverse models ¹.

1 Introduction

The widespread application of DNNs in critical domains, such as self-driving cars (Bose et al., 2021) and medical diagnosis (Gongye et al., 2020), has emphasized the need to study their reliability and performance in noisy environments. Also, to improve their performance, custom hardware accelerators have emerged as a promising solution given the high computational demands of DNNs. However, it is essential to ensure that DNNs retain their efficiency and reliability when deployed on specialized hardware with a limited energy budget and in the presence of hardware faults. Although quantizing and compressing DNN weights can improve robustness and decrease the potential impact of bit-flip errors (Qian et al., 2023), adversarial attacks can still cause severe performance deterioration with only a few vulnerable bits (Rakin et al., 2019).

Bit-flips in memories can occur randomly as a result of undesirable events such as chip fabrication variations, supply voltage variations, or targeted attacks. The row-hammer attack (RHA), where multiple accesses can disturb DRAM rows and cause bit-flips in other rows, has been extensively researched (Kim et al., 2014), with DRAMs being highly susceptible to this type of attack due to their dense architecture. Furthermore,

¹Code is available at <https://github.com/kmchiti/WCAT>